

TraceBridge AI: Comprehensive Data Dictionary & Schema Definition

Version: 2.0 (Detailed Engineering Specification)
Target Audience: BU Spark Data Science & Engineering Teams, TraceBridge Core Contributors
Database Engine: Google Cloud Firestore (NoSQL Document Store)
Vector Engine: Vertex AI & Firestore Native Vector Search

1. Executive Summary

This document provides a highly detailed schema definition for all data entities within the TraceBridge AI ecosystem. It defines the exact JSON structure, field-level types, cardinality, and referential integrity mechanisms used across both the Regulatory Intelligence Engine (Module 1) and the Post-Market Surveillance Engine (Module 3).

2. Module 1: RAG & 510(k) Intelligence Entities

2.1 knowledgeBase Collection

Purpose: Stores the vectorized corpus of historical FDA precedents, ISO standards, and predicate device summaries. This is the core "ground truth" retrieved during RAG operations.
Primary Key: documentId_chunkIndex

Field Name	Data Type	Required	Description / Allowed Values
text_content	String	Yes	The raw chunk text extracted from the regulatory source. Max 2,000 chars.
source_file	String	Yes	The filename or URI of the original source (e.g., FDA_Guidance_Cybersecurity_2023.pdf).
k_number	String	No	The 510(k) clearance number if the document is a predicate summary (e.g., K192482).
device_name	String	No	Extracted device trade name from the precedent.
regulatory_standard	String	No	If derived from a standard, the exact standard ID (e.g., ISO 14971:2019).
embedding	Vector<768>	Yes	768-dimensional float32 vector generated via gemini-embedding-2.
created_at	Timestamp	Yes	System timestamp of ingestion.

2.2 uploads Collection

Purpose: Represents a single "Submission Readiness" project created by an engineering user. It acts as the parent object for user-uploaded engineering documents and the resulting gap analysis.
Primary Key: uploadId (UUID v4)

Field Name	Data Type	Required	Description / Allowed Values
userId	String	Yes	The Firebase Auth UID of the tenant/user who initiated the upload.
			Human-readable name for the submission (e.g., Horizon POD)

projectName	String	Yes	510(k)).
productCode	String	No	FDA 3-letter product code (e.g., MKJ for continuous glucose monitors).
standards	Array<String>	Yes	List of standards applied for the gap analysis (e.g., ["ISO 14971", "IEC 62304"])).
status	String	Yes	pending_upload, processing_embeddings, analyzing, complete, failed.
createdAt	Timestamp	Yes	When the project was initiated.

2.3 documents Collection (Sub-Entity of Uploads)

Purpose: Metadata and text extraction results for individual files uploaded to a specific project.

Primary Key: documentId (UUID v4)

Foreign Key: uploadId -> uploads.uploadId

Field Name Data Type Required Description / Allowed Values

uploadId	String	Yes	References the parent upload project.
fileName	String	Yes	Original uploaded filename (e.g., Risk_Management_Plan.pdf).
mimeType	String	Yes	application/pdf, text/plain, application/vnd.openxmlformats...
sizeBytes	Number	Yes	File size in bytes for audit limits.
status	String	Yes	pending, chunking, embedded, failed.

2.4 gapResults Collection

Purpose: The structured output of the Gemini AI gap analysis, mapping FDA requirements against the user's uploaded documents.

Primary Key: gapId (UUID v4)

Foreign Key: uploadId -> uploads.uploadId

Field Name	Data Type	Required	Description / Allowed Values
uploadId	String	Yes	References the parent upload project.
standard	String	Yes	The regulatory standard being evaluated (e.g., FDA Cybersecurity 2023).
section	String	Yes	The specific section or clause (e.g., Section 4.1.2).
requirement	String	Yes	The plaintext regulatory requirement being tested.
status	String	Yes	compliant (Passed), gap_detected (Failed/Missing), needs_review (Ambiguous).
confidence	Number	Yes	AI confidence score (0-100).
missingRequirement	String	No	Populated if status == gap_detected. Explains exactly what the engineers forgot to include.
citations	Array<Object>	No	Array containing { source: String, quote: String, section: String }. Proof of compliance if status == compliant.

3. Module 3: Post-Market Surveillance Entities

3.1 complaints Collection

Purpose: The central repository for all raw, incoming adverse events and their subsequent AI classifications.
Primary Key: complaintId (UUID v4)

Field Name	Data Type	Required	Description / Allowed Values
raw_text	String	Yes	Unstructured narrative text of the adverse event.
source	String	Yes	Ingestion channel: email, webhook, manual_entry.
status	String	Yes	pending_classification, classified, failed.
received_at	Timestamp	Yes	Ingestion timestamp.
device_component	String	No	Extracted via Gemini (e.g., Battery, Motor, Software).
failure_type	String	No	Extracted failure mode (e.g., Premature Drain, Unresponsive UI).
severity	String	No	critical (Death/Injury risk), major (Device failure), minor (Cosmetic/Nuance).
needs_mdr	Boolean	No	Automatically set to true if AI determines FDA 30-day reporting applies.

3.2 alerts Collection

Purpose: Generated automatically by backend CRON jobs. Used to signal critical issues (like "Systemic Trend Alerts") to the executive dashboard.
Primary Key: alertId (UUID v4)

Field Name	Data Type	Required	Description / Allowed Values
type	String	Yes	safety_signal (Single severe event), systemic_trend (Statistically significant cluster).
title	String	Yes	Brief dashboard title (e.g., Systemic Trend: Battery Premature Drain).
severity	String	Yes	high, medium, low. Influences UI rendering (Red vs Yellow).
status	String	Yes	open, acknowledged (A human clicked it), closed.
message	String	Yes	Detailed AI synthesis of the issue and recommended regulatory action.
detected_at	Timestamp	Yes	When the CRON job flagged the anomaly.

4. Evaluation Dataset Metadata (annotated_outcomes.json)

The Golden Dataset uses a lightweight JSON structure specifically for precision/recall testing.

Field Name	Data Type	Required	Description
filename	String	Yes	The mock document being tested (e.g., Risk_Management_Plan_DRAFT.txt).
rule_id	String	Yes	The specific FDA/ISO rule being tested.
requirement	String	Yes	The text of the rule.
expected_found	Boolean	Yes	Ground Truth. If true, the document contains the evidence. If false, the document lacks it.
ground_truth_reasoning	String	Yes	Justification for why the human grader assigned true/false. Used to debug AI hallucination.